

Module 1: Git Basics & Repository Setup

Assignment 1: New Project Setup

Scenario:

You are starting a new Java project for a hospital system.

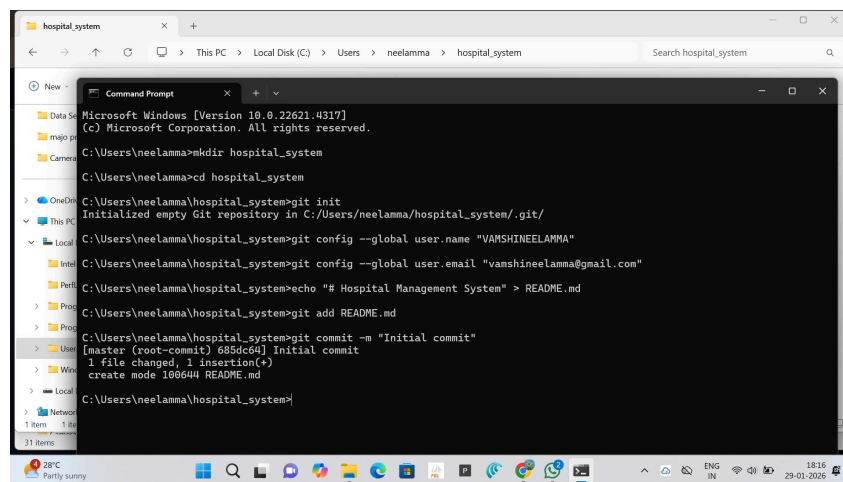
Task:

Create a new Git repository

Configure your username and email

Add a README.md

Make the first commit



```
Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\neelamma>mkdir hospital_system

C:\Users\neelamma>cd hospital_system

C:\Users\neelamma\hospital_system>git init
Initialized empty Git repository in C:/Users/neelamma/hospital_system/.git/

C:\Users\neelamma\hospital_system>git config --global user.name "VAMSHINEELAMMA"

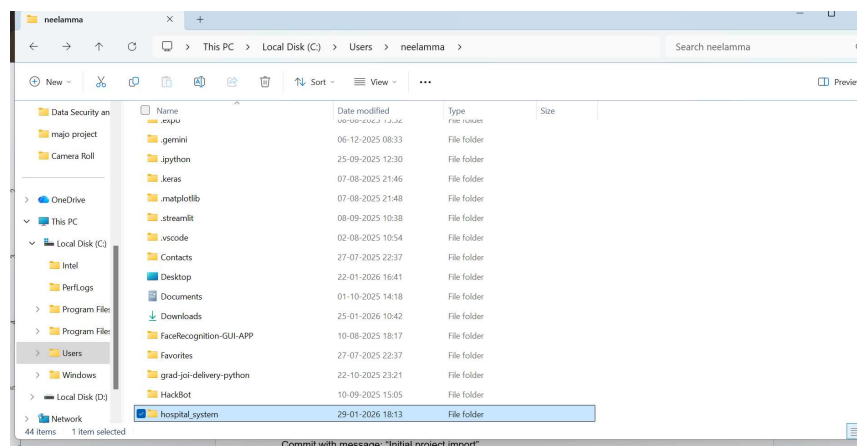
C:\Users\neelamma\hospital_system>git config --global user.email "vamshineelamma@gmail.com"

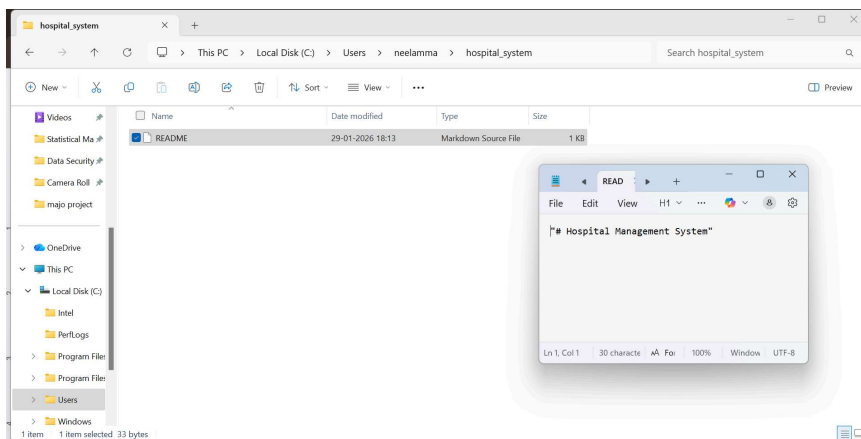
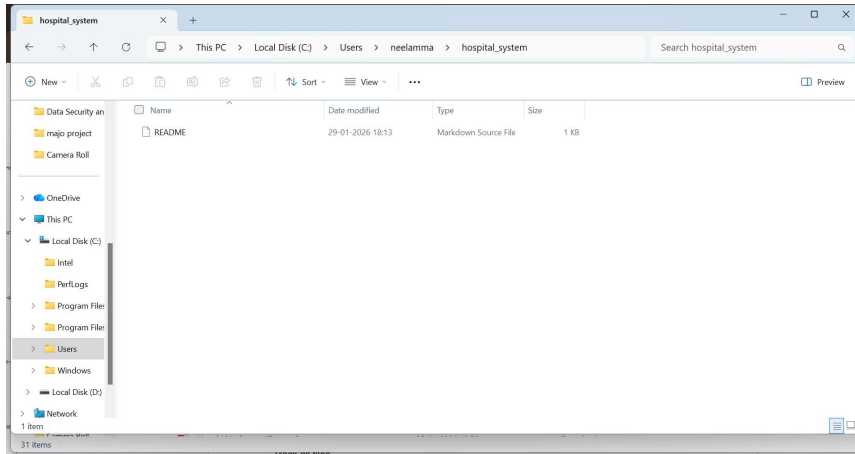
C:\Users\neelamma\hospital_system>echo "# Hospital Management System" > README.md

C:\Users\neelamma\hospital_system>git add README.md

C:\Users\neelamma\hospital_system>git commit -m "Initial commit"
[master (root-commit) 685dc64] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 README.md

C:\Users\neelamma\hospital_system>
```





Assignment 2: Existing Project to Git

Scenario:

You received a folder containing an existing project without version control.

Task:

Initialize Git in the folder

Track all files

Commit with message: "Initial project import"

```

C:\Users\neelamma\hospital_system>git init
Initialized empty Git repository in C:/Users/neelamma/hospital_system/.git/
C:\Users\neelamma\hospital_system>git config --global user.name "VAMSHINEELAMMA"
C:\Users\neelamma\hospital_system>git config --global user.email "vamshineelamma@gmail.com"
C:\Users\neelamma\hospital_system>echo "# Hospital Management System" > README.md
C:\Users\neelamma\hospital_system>git add README.md
C:\Users\neelamma\hospital_system>git commit -m "Initial commit"
[master (root-commit) 685dc64] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 README.md
C:\Users\neelamma\hospital_system>git status
On branch master
nothing to commit, working tree clean
C:\Users\neelamma\hospital_system>git log --oneline
685dc64 (HEAD -> master) Initial commit
C:\Users\neelamma\hospital_system>git init
Reinitialized existing Git repository in C:/Users/neelamma/hospital_system/.git/
C:\Users\neelamma\hospital_system>git add .
C:\Users\neelamma\hospital_system>git commit -m "Initial project import"
On branch master
nothing to commit, working tree clean
C:\Users\neelamma\hospital_system>

```

Assignment 3: Ignoring Unwanted Files

Scenario:

Your project generates .log, .class, and node_modules/ files.

Task:

Create .gitignore

Ignore those files

Verify ignored files are not tracked

```
MINGW64/C:/Users/heelamma/hospital-system
$ cd hospital-system
$ ls -a
. .git/ .gitignore README.md
$ touch .gitignore
$ ls
. .git/ .gitignore README.md
$ ls
. .git/ .gitignore README.md
$ ls -a
. .git/ .gitignore README.md
$ nano .gitignore
$ ls -a
. .git/ .gitignore README.md
$ nano .gitignore
```

```
GNU nano 2.9.5 .gitignore
node_modules/
*.log
```

```
MINGW64/C:/Users/heelamha/hospital-system
/ ./ ./ .git/ .gitignore README.md
root@helamha02240024 MINGW64 ~/hospital-system (master)
$ nano .gitignore
root@helamha02240024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
no commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   README.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .gitignore

root@helamha02240024 MINGW64 ~/hospital-system (master)
$ git add .gitignore
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time git touches it
root@helamha02240024 MINGW64 ~/hospital-system (master)
$ git add .
root@helamha02240024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
no commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   .gitignore
    new file:   README.md

root@helamha02240024 MINGW64 ~/hospital-system (master)
$ git commit -m "Initial commit with README and gitignore"
[master (root-commit) 602c41f] Initial commit with README and gitignore
 2 files changed, 4 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 README.md

root@helamha02240024 MINGW64 ~/hospital-system (master)
$ !
```

Assignment 4: Check Repository Status

Scenario:

Before committing, you want to confirm what has changed.

Task:

Modify two files

Check repository status

Identify staged vs unstaged files

```
MINGW64/C:/Users/heelamha/hospital-system
new File: README.md
root@helamha02240024 MINGW64 ~/hospital-system (master)
$ git commit -m "Initial commit with README and gitignore"
[master (root-commit) 602c41f] Initial commit with README and gitignore
 2 files changed, 4 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 README.md

root@helamha02240024 MINGW64 ~/hospital-system (master)
$ nano README.md
root@helamha02240024 MINGW64 ~/hospital-system (master)
$ nano notes.txt
root@helamha02240024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore --staged <file>..." to discard changes in working directory)
    modified:   README.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    notes.txt

no changes added to commit (use "git add" and/or "git commit -a")

root@helamha02240024 MINGW64 ~/hospital-system (master)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time git touches it
root@helamha02240024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   README.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    notes.txt

root@helamha02240024 MINGW64 ~/hospital-system (master)
$
```

```
MINGW64/C:/Users/heelamha/hospital-system
root@helamha02240024 MINGW64 ~/hospital-system (master)
$ nano notes.txt
root@helamha02240024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore --staged <file>..." to discard changes in working directory)
    modified:   README.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    notes.txt

no changes added to commit (use "git add" and/or "git commit -a")

root@helamha02240024 MINGW64 ~/hospital-system (master)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time git touches it
root@helamha02240024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   README.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    notes.txt

root@helamha02240024 MINGW64 ~/hospital-system (master)
$ git add .
warning: in the working copy of 'notes.txt', LF will be replaced by CRLF the next time git touches it
root@helamha02240024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   README.md
    new file:   notes.txt

root@helamha02240024 MINGW64 ~/hospital-system (master)
$
```

Module 2: Staging & Commit Management

Assignment 5: Selective Commit

Scenario:

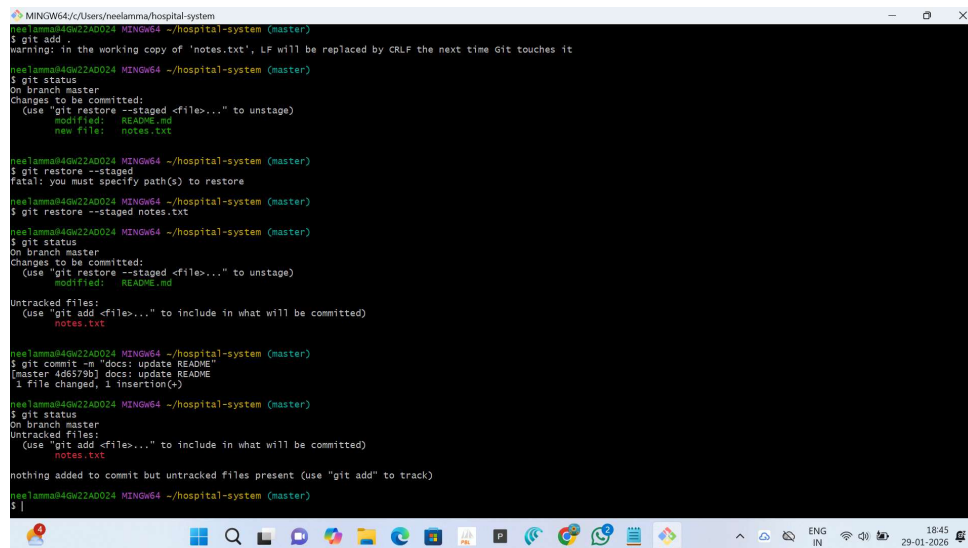
You modified 3 files but want to commit only 1.

Task:

Stage only one file

Commit it

Leave others unstaged



```
neelamma@49w22AD024 MINGW64 ~/hospital-system (master)
$ git add
warning: in the working copy of 'notes.txt', LF will be replaced by CRLF the next time Git touches it
neelamma@49w22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md
        new file:   notes.txt

neelamma@49w22AD024 MINGW64 ~/hospital-system (master)
$ git restore --staged
fatal: you must specify path(s) to restore
neelamma@49w22AD024 MINGW64 ~/hospital-system (master)
$ git restore --staged notes.txt
neelamma@49w22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        notes.txt

neelamma@49w22AD024 MINGW64 ~/hospital-system (master)
$ git commit -m "docs: update README"
[master 4de579b] docs: update README
1 file changed, 1 insertion(+)

neelamma@49w22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        notes.txt

nothing added to commit but untracked files present (use "git add" to track)
neelamma@49w22AD024 MINGW64 ~/hospital-system (master)
$
```

Assignment 6: Commit Message Standards

Scenario:

Your team enforces meaningful commit messages.

Task:

Make a commit with a clear, descriptive message

Follow this format: type: description

```
MINGW64~/c:/Users/nelamma/hospital-system
Untracked Files:
(use "git add <file>..." to include in what will be committed)
notes.txt

nelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git commit -m "docs: update README"
[master 4d6579b] docs: update README
1 file changed, 1 insertion(+)

nelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Untracked Files:
(use "git add <file>..." to include in what will be committed)
notes.txt

nothing added to commit but untracked files present (use "git add" to track)

nelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ nano notes.txt
$ git add notes.txt
warning: in the working copy of 'notes.txt', LF will be replaced by CRLF the next time Git touches it

nelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Changes to be committed:
(use "git restore --staged <file>..." to unstage)
New File:   notes.txt

nelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git commit -m "docs: adding more information"
[master 8b069e8] docs: adding more information
1 file changed, 2 insertions(+)
create mode 100644 notes.txt

nelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git log --oneline
8b069e8 (HEAD -> master) docs: adding more information
4d6579b docs: update README
352c41f Initial commit with README and gitignore

nelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$
```

Assignment 7: Amend Last Commit

Scenario:

You forgot to add one file to your last commit.

Task:

Add the missing file

Amend the previous commit (without creating a new one)

```
notes.txt

nelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git add .
warning: in the working copy of 'notes.txt', LF will be replaced by CRLF the next time Git touches it

nelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Changes to be committed:
(use "git restore --staged <file>..." to unstage)
modified:   README.md
new file:   notes.txt

nelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git restore --staged
fatal: you must specify path(s) to restore

nelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git restore --staged notes.txt

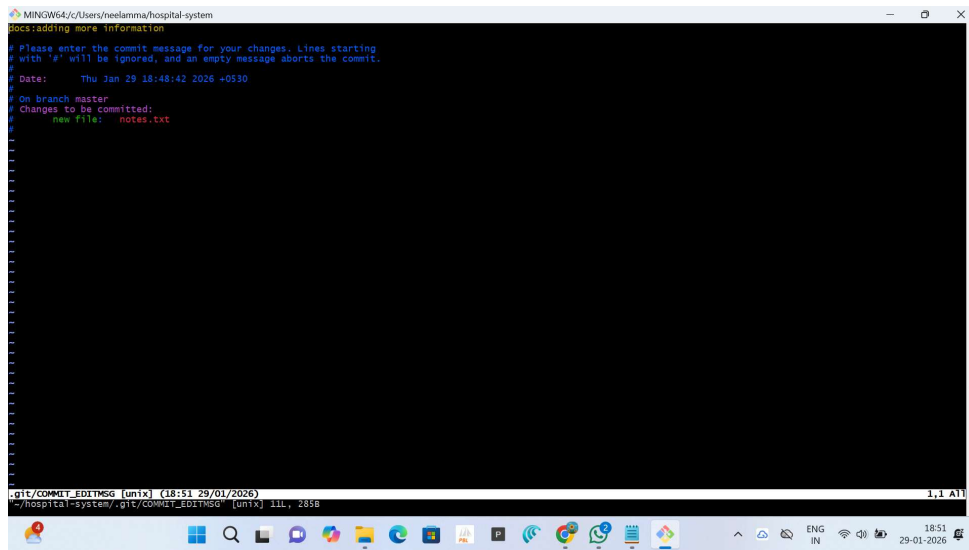
nelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Changes to be committed:
(use "git restore --staged <file>..." to unstage)
modified:   README.md

Untracked files:
(use "git add <file>..." to include in what will be committed)
notes.txt

nelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git commit -m "docs: update README"
[master 4d6579b] docs: update README
1 file changed, 1 insertion(+)

nelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Untracked files:
(use "git add <file>..." to include in what will be committed)
notes.txt

docs: adding more information
```



Assignment 8: View Commit History

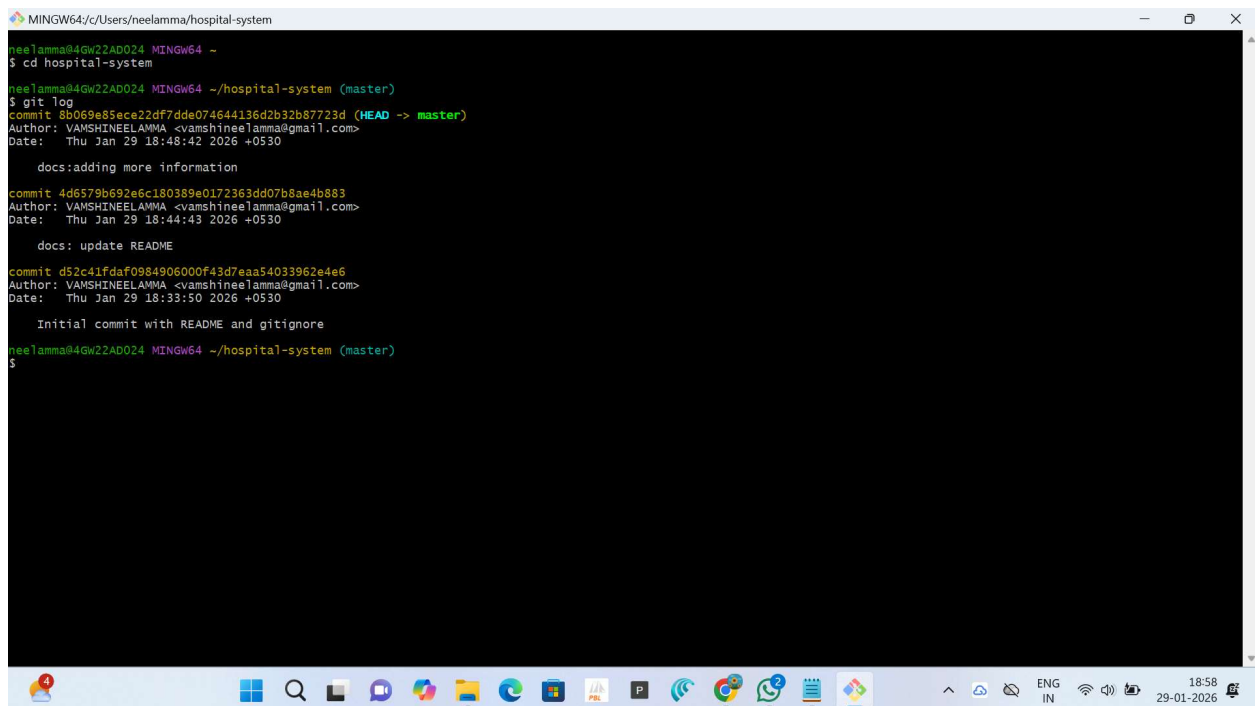
Scenario:

Your manager wants to review previous changes.

Task:

[View commit history](#)

Display commits in one-line format



Module 3: Branching & Merging

Assignment 9: Feature Development

Scenario:

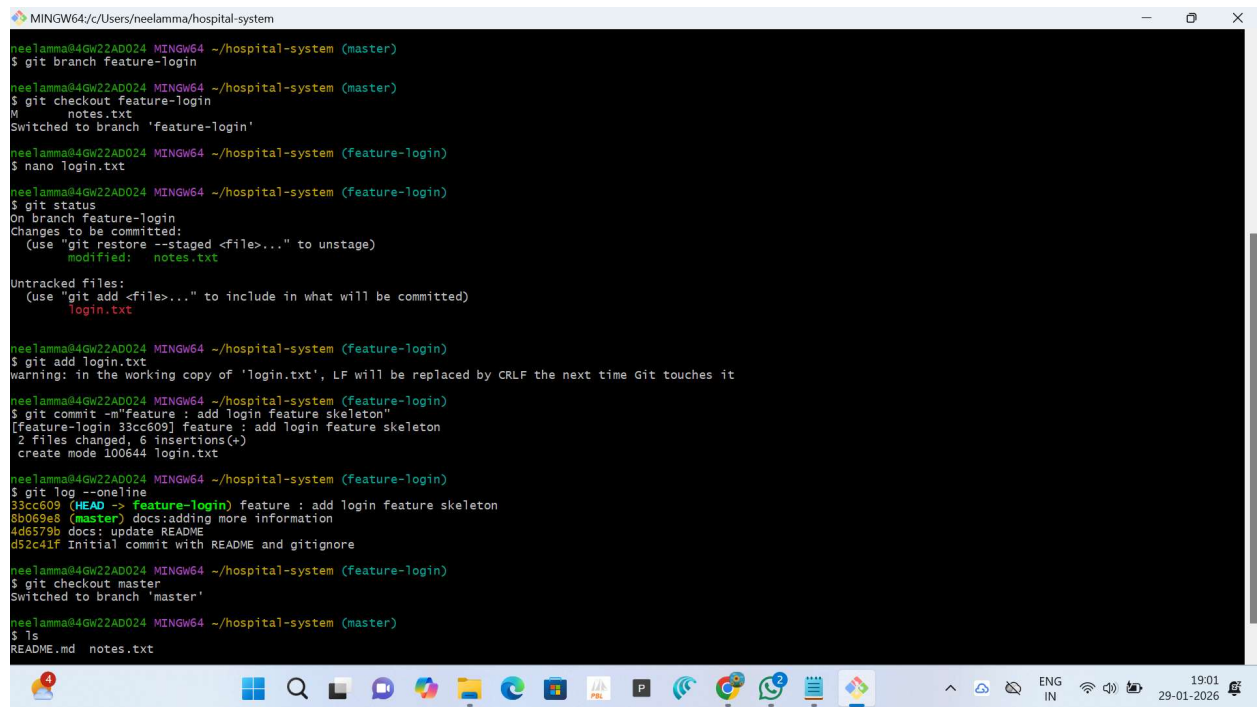
You are working on a new feature called login.

Task:

Create a branch feature-login

Switch to it

Make changes and commit



```
MINGW64/c/Users/neelamma/hospital-system
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git branch feature-login
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git checkout feature-login
M   notes.txt
Switched to branch 'feature-login'
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login)
$ nano login.txt
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login)
$ git status
On branch feature-login
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   notes.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        login.txt

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login)
$ git add login.txt
warning: in the working copy of 'login.txt', LF will be replaced by CRLF the next time Git touches it
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login)
$ git commit -m"feature : add login feature skeleton"
[feature-login 33cc609] feature : add login feature skeleton
 2 files changed, 6 insertions(+)
 create mode 100644 login.txt

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login)
$ git log --oneline
33cc609 (HEAD -> feature-login) feature : add login feature skeleton
5b069e8 (master) docs:adding more information
4d6979b docs: update README
452c41f Initial commit with README and gitignore

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login)
$ git checkout master
Switched to branch 'master'

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ ls
README.md  notes.txt
```

Assignment 10: Switch Between Branches

Scenario:

You need to move back to the main branch urgently.

Task:

List branches

Switch to main

Verify active branch


```
MINGW64~/c/Users/neelamma/hospital-system
(use "git restore --staged <file>..." to unstage)
modified: notes.txt

Untracked files:
(use "git add <file>..." to include in what will be committed)
login.txt

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login)
$ git add login.txt
warning: in the working copy of 'login.txt', LF will be replaced by CRLF the next time Git touches it

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login)
$ git commit -m"feature : add login feature skeleton"
[feature-login 33cc609] feature : add login feature skeleton
2 files changed, 6 insertions(+)
create mode 100644 login.txt

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login)
$ git log --oneline
33cc609 (HEAD -> feature-login) feature : add login feature skeleton
8b069e8 (master) docs:adding more information
4d6979b docs: update README
d52c41f Initial commit with README and gitignore

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login)
$ git checkout master
Switched to branch 'master'

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ ls
README.md notes.txt

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git branch
* feature-login
* master

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git checkout feature-login
Switched to branch 'feature-login'

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login)
$ git checkout master
Switched to branch 'master'

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$
```

Assignment 11: Merge Feature Branch

Scenario:

The feature-login is complete and tested.

Task:

Merge feature-login into main

Delete the feature branch after merge

```
MINGW64/c/Users/neelamma/hospital-system

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login)
$ git checkout master
Switched to branch 'master'

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ ls
README.md  notes.txt

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git branch
* master
$ git checkout feature-login
Switched to branch 'feature-login'

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login)
$ git checkout master
Switched to branch 'master'

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git checkout master
Already on 'master'

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git merge feature-login
Updating 8b069e8..33cc609
Fast-forward
 login.txt | 5 +++++
 notes.txt | 1 +
 2 files changed, 6 insertions(+)
 create mode 100644 login.txt

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git log --oneline
33cc609 (HEAD -> master, feature-login) feature : add login feature skeleton
8b069e8 docs: adding more information
4d6579b docs: update README
d52c41f Initial commit with README and gitignore

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ ls
README.md  login.txt  notes.txt

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ |
```

Assignment 12: Merge Conflict Handling

Scenario:

Two developers edited the same line in the same file.

Task:

Create a merge conflict

Resolve it manually

Complete the merge

```
MINGW64/c/Users/neelamma/hospital-system

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ ls
README.md  login.txt  notes.txt

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git branch -d feature-login
Deleted branch feature-login (was 33cc609).

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git checkout master
Already on 'master'

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git branch
* master
$ nano README.md

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git commit -m"docs: updated README"
On branch master
nothing to commit, working tree clean

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git checkout feature-login
error: pathspec 'feature-login' did not match any file(s) known to git

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git branch
* master
$ git branch feature-login-v1

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git checkout feature-login-v1
Switched to branch 'feature-login-v1'

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ git branch
* feature-login-v1
  master
```

```
MINGW64/c/Users/neelamma/hospital-system
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ nano README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ git add README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ git commit -m "docs : update README for feature-login-v1"
On branch feature-login-v1
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    README.md

nothing added to commit but untracked files present (use "git add" to track)
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ nano README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ git commit -m "docs : update README for feature-login-v1"
On branch feature-login-v1
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    README.md

nothing added to commit but untracked files present (use "git add" to track)
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ cls
bash: cls: command not found
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ ls
README.md login.txt notes.txt
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ git status
On branch feature-login-v1
nothing to commit, working tree clean
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ git add README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
```

```
MINGW64/c/Users/neelamma/hospital-system
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ git commit -m "docs : update README for feature-login-v1"
On branch feature-login-v1
nothing to commit, working tree clean
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ git status
On branch feature-login-v1
nothing to commit, working tree clean
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ nano README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ git commit -m "docs : update README for feature-login-v1"
On branch feature-login-v1
nothing to commit, working tree clean
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ git checkout master
git: 'checkout' is not a git command. See 'git --help'.

The most similar command is
  checkout
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ git checkout master nano README.md
git: 'checkout' is not a git command. See 'git --help'.

The most similar command is
  checkout
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ git log --oneline -1
33cc609 (HEAD -> feature-login-v1, master) feature : add login feature skeleton
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ git checkout master
Switched to branch 'master'
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git branch
  feature-login-v1
```

```
MINGW64/c/Users/neelamma/hospital-system
master
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git checkout feature-login-v1
git: 'checkout' is not a git command. See 'git --help'.

The most similar command is
  checkout

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git checkout feature-login-v1
Switched to branch 'feature-login-v1'

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (feature-login-v1)
$ nano README.md
Merge branch 'feature-login-v1'
# Please enter a commit message to explain why this merge is necessary,
# especially if it merges an updated upstream into a topic branch.
#
# Lines starting with '#' will be ignored, and an empty message aborts
# the commit.

```

```
hint: waiting for your editor to close the file...

[master 38304ac] Merge branch 'feature-login-v1'
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
nothing to commit, working tree clean
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git log --oneline
38304ac (HEAD -> master) Merge branch 'feature-login-v1'
48a86d7 docs : update README for master
84c09f4 (feature-login-v1) docs : update README for feature-login-v1
33cc609 feature : add login feature skeleton
3b069e8 docs:adding more information
4d6979b docs: update README
d52c41f Initial commit with README and gitignore
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$
```

Module 4: Undoing & Recovery

Assignment 13: Undo File Changes

Scenario:

You edited a file but want to discard the changes.

Task:

Restore the file to last committed state

```
MINGW64/c/Users/neelamma/hospital-system
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ nano README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git diff
diff --git a/README.md b/README.md
index b926859..b2be321 100644
--- a/README.md
+++ b/README.md
@@ -1,3 +1,5 @@
#Hospital Management System
A project that tracks patients and keeps them updated.
keep going
+temporary change for assignment 14
+A wrong change

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git restore README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
nothing to commit, working tree clean

neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ nano README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$
```

Assignment 14: Unstage a File

Scenario:

You accidentally staged a file.

Task:

Remove it from staging

Keep the changes in working directory

```
MINGW64/c/Users/neelamma/hospital-system
GNU nano 8.5
#Hospital Management System
A project that tracks patients and keeps them updated.
Keep going
temporary change for assignment 14

[ Soft wrapping of overlong lines enabled ]
AG Help      AO Write Out  AF Where Is  AK Cut       AT Execute   AC Location  M-U Undo    M-A Set Mark M-J To Bracket M-B Previous
AX Exit      AR Read File  \ Replace    AU Paste     AJ Justify   ^/ Go To Line M-E Redo    M-6 Copy     AB Where Was  M-F Next
```

```
MINGW64/c/Users/neelamma/hospital-system
".git/COMMIT_EDITMSG" [unix] 19L, 6018 written
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git log --oneline
38304ac (HEAD -> master) Merge branch 'feature-login-v1'
48a86d7 docs : update README for master
84c09f4 (feature-login-v1) docs : update README for feature-login-v1
33cc609 feature : add login feature skeleton
9b069e8 docs:adding more information
4d6579b docs: update README
d52c41f Initial commit with README and gitignore
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git branch
  feature-login-v1
* master
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ nano README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git add README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ nano README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git restore --staged README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md
no changes added to commit (use "git add" and/or "git commit -a")
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$
```

Assignment 15: Reset to Previous Commit

Scenario:

Your last commit broke the application.

Task:

Reset the repository to the previous commit

Keep changes locally

```
MINGW64/c/Users/neelamma/hospital-system
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ nano README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ nano README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git add README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git commit -m"temp:assignment 15 text commit"
[master 69e8b57] temp:assignment 15 text commit
1 file changed, 1 insertion(+)
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git log --oneline -1
69e8b57 (HEAD -> master) temp:assignment 15 text commit
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git reset --soft HEAD~1
fatal: ambiguous argument 'HEAD~1': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git reset --soft HEAD~1
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git log --oneline
38304ac (HEAD -> master) Merge branch 'feature-login-v1'
48a86d7 docs : update README for master
34c09f4 (feature-login-v1) docs : update README for feature-login-v1
33cc609 feature : add login feature skeleton
8b069e8 docs:adding more information
4d6579b docs: update README
d52c41f Initial commit with README and gitignore
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ |
```

Assignment 16: Delete a Commit Completely

Scenario:

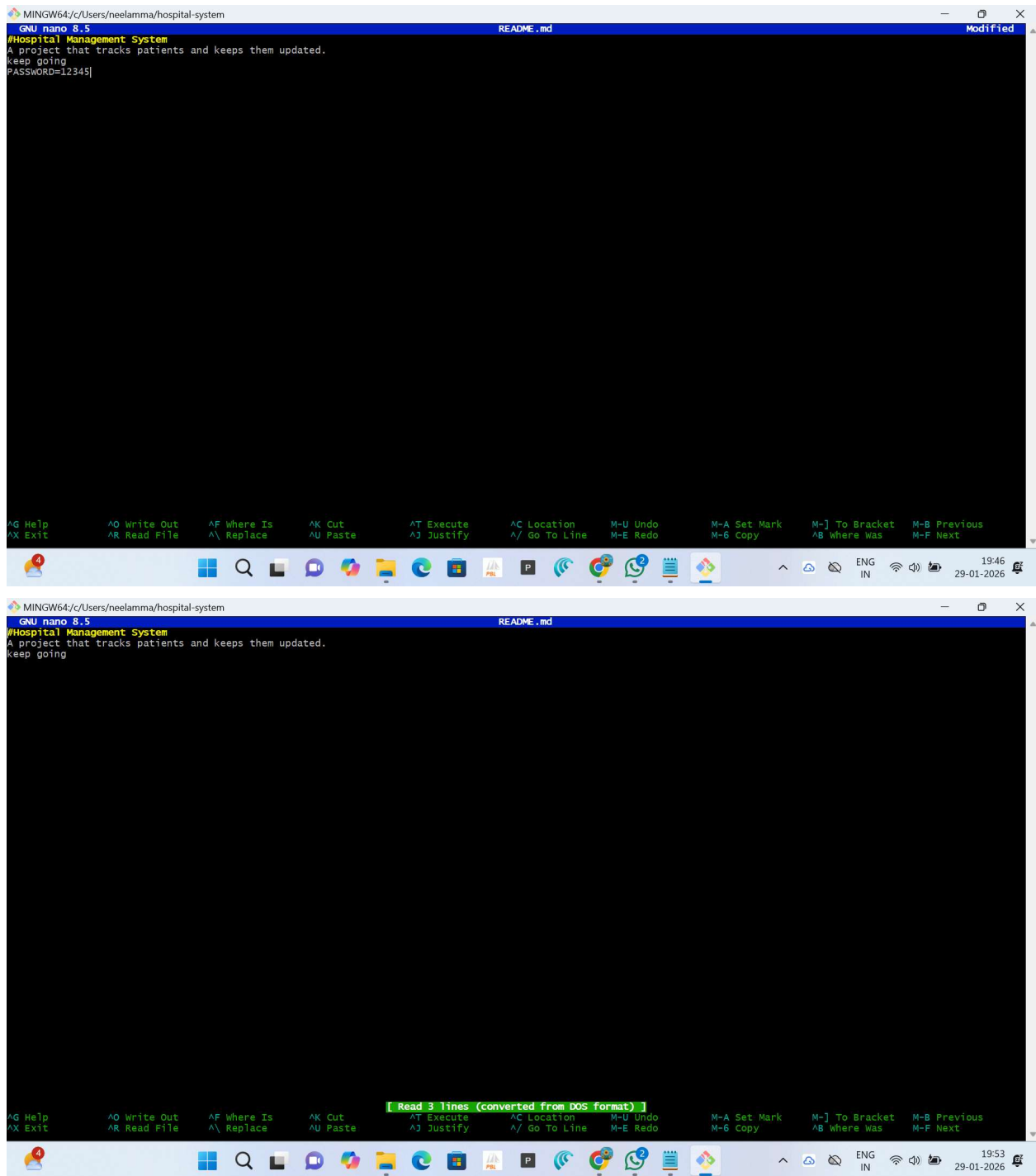
You committed sensitive data by mistake.

Task:

Remove the commit permanently

Ensure it does not appear in history


```
MINGW64:/c/Users/neelamma/hospital-system
d52c41f Initial commit with README and gitignore
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git reset --hard
HEAD is now at 38304ac Merge branch 'feature-login-v1'
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
nothing to commit, working tree clean
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ nano README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git add README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git commit -m "bad:accidentally committed password"
[master 958f8e1] bad:accidentally committed password
1 file changed, 1 insertion(+)
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git log --oneline -1
958f8e1 (HEAD -> master) bad:accidentally committed password
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git reset --hard HEAD~1
fatal: cannot do hard reset with paths.
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git reset --hard HEAD~1
HEAD is now at 38304ac Merge branch 'feature-login-v1'
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git log --oneline -1
38304ac (HEAD -> master) Merge branch 'feature-login-v1'
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
nothing to commit, working tree clean
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ nano README.md
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ |
```

● Module 5: Remote Repositories (GitHub/GitLab)

Assignment 17: Connect Local Repo to Remote

Scenario:

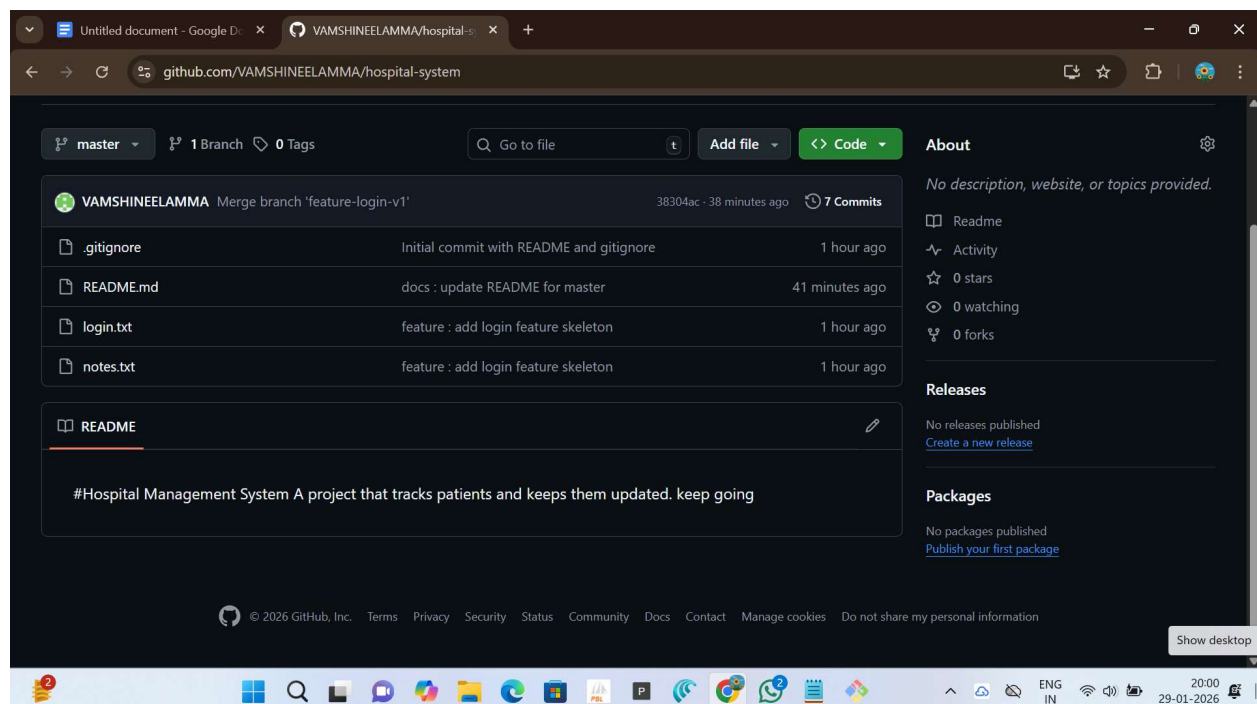
Your project must be pushed to a remote repository.

Task:

Add remote origin

Push local commits

```
MINGW64/c/Users/neelamma/hospital-system
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git remote add origin https://github.com/VAMSHINEELAMMA/hospital-system.git
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git remote -v
origin  https://github.com/VAMSHINEELAMMA/hospital-system.git (fetch)
origin  https://github.com/VAMSHINEELAMMA/hospital-system.git (push)
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git push -u origin master
Enumerating objects: 19, done.
Counting objects: 100% (19/19), done.
Delta compression using up to 4 threads
Compressing objects: 100% (17/17), done.
Writing objects: 100% (19/19), 1.82 KiB | 466.00 KiB/s, done.
Total 19 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (6/6), done.
To https://github.com/VAMSHINEELAMMA/hospital-system.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
neelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$
```



Assignment 18: Clone a Repository

Scenario:

You joined a team and need the existing codebase.

Task:

Clone a remote repository
Check branch list

```
MINGW64: c:/Users/neeelamma/Desktop/hospital-system
neeelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ pwd
/c:/Users/neeelamma/hospital-system
neeelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
neeelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ cd Desktop
bash: cd: Desktop: No such file or directory
neeelamma@4Gw22AD024 MINGW64 ~/hospital-system (master)
$ cd ..
neeelamma@4Gw22AD024 MINGW64 ~
$ cd /c:/Users/neeelamma/Desktop
neeelamma@4Gw22AD024 MINGW64 ~/Desktop
$ git clone https://github.com/VAMSHINEELAMMA/hospital-system.git
Cloning into 'hospital-system'...
remote: Enumerating objects: 19, done.
remote: Counting objects: 100% (19/19), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 19 (delta 6), reused 19 (delta 6), pack-reused 0 (from 0)
Receiving objects: 100% (19/19), done.
Resolving deltas: 100% (6/6), done.
neeelamma@4Gw22AD024 MINGW64 ~/Desktop
$ cd hospital-system
neeelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ pwd
/c:/Users/neeelamma/Desktop/hospital-system
neeelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
neeelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ |
```

Assignment 19: Push Changes

Scenario:

You completed a bug fix.

Task:

Commit changes

Push to remote branch

```
MINGW64/c/Users/neelamma/Desktop/hospital-system
neelamma@4Gw22AD024 MINGW64 ~/Desktop
$ git clone https://github.com/VAMSHINEELAMMA/hospital-system.git
Cloning into 'hospital-system'...
remote: Enumerating objects: 19, done.
remote: Counting objects: 100% (19/19), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 19 (delta 6), reused 19 (delta 6), pack-reused 0 (from 0)
Receiving objects: 100% (19/19), done.
Resolving deltas: 100% (6/6), done.
neelamma@4Gw22AD024 MINGW64 ~/Desktop
$ cd hospital-system
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ pwd
/c/Users/neelamma/Desktop/hospital-system
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ pwd
/c/Users/neelamma/Desktop/hospital-system
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git remote -v
origin https://github.com/VAMSHINEELAMMA/hospital-system.git (fetch)
origin https://github.com/VAMSHINEELAMMA/hospital-system.git (push)
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git pull origin master
From https://github.com/VAMSHINEELAMMA/hospital-system
 * branch      master       -> FETCH_HEAD
Already up to date.
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$
```

Assignment 20: Pull Latest Changes

Scenario:

Your teammate pushed updates.

Task:

Pull changes

Verify updated files

```
MINGW64/c/Users/neelamma/Desktop/hospital-system
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ pwd
/c/Users/neelamma/Desktop/hospital-system
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git remote -v
origin https://github.com/VAMSHINEELAMMA/hospital-system.git (fetch)
origin https://github.com/VAMSHINEELAMMA/hospital-system.git (push)
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git pull origin master
From https://github.com/VAMSHINEELAMMA/hospital-system
 * branch      master       -> FETCH_HEAD
Already up to date.
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git tag -a v1.0 -m "version 1.0"
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git tag
v1.0
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git push origin v1.0
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 177 bytes | 177.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/VAMSHINEELAMMA/hospital-system.git
 * [new tag]      v1.0 -> v1.0
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git push --tags
Everything up-to-date
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ |
```

Assignment 21: Resolve Pull Conflicts

Scenario:

Pulling changes caused a conflict.

Task:

Resolve conflict

Commit resolved version

```
MINGW64~/Users/neelamma/Desktop/hospital-system
$ git tag
v1.0
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git push origin v1.0
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 177 bytes | 177.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/VAMSHINEELAMMA/hospital-system.git
 * [new tag]         v1.0 -> v1.0
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git push --tags
Everything up-to-date
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git pull origin master
From https://github.com/VAMSHINEELAMMA/hospital-system
 * branch            master       -> FETCH_HEAD
Already up to date.
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git add README.md
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git commit -m "Resolve merge conflict"
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ |
```

Module 6: Advanced Git Usage

Assignment 22: Stash Work

Scenario:

You are halfway through work but need to switch branches.

Task:

Stash changes

Switch branch

Apply stash later

```
MINGW64/c/Users/neelamma/Desktop/hospital-system
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ nano README.md

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git stash
Saved working directory and index state WIP on master: 38304ac Merge branch 'feature-login-v1'

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git stash list
stash@{0}: WIP on master: 38304ac Merge branch 'feature-login-v1'

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git stash apply
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git stash drop
Dropped refs/stash@{0} (a51cc83781af2cb1e78b9878c2ca2cd0225b8bf4)

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$
```

Assignment 23: View Differences

Scenario:

You want to review changes before committing.

Task:

View differences between working tree and last commit

```
MINGW64/c/Users/neelamma/Desktop/hospital-system
Dropped refs/stash@{0} (a51cc83781af2cb1e78b9878c2ca2cd0225b8bf4)

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ nano README.md

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git diff
diff --git a/README.md b/README.md
index b926859..dea1994 100644
--- a/README.md
+++ b/README.md
@@ -1,3 +1,5 @@
 #Hospital Management System
 A project that tracks patients and keeps them updated.
 keep going
+modified
+differec4 gor assignment 23

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git diff README.md
diff --git a/README.md b/README.md
index b926859..dea1994 100644
--- a/README.md
+++ b/README.md
@@ -1,3 +1,5 @@
 #Hospital Management System
 A project that tracks patients and keeps them updated.
 keep going
+modified
+differec4 gor assignment 23

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git diff --staged

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$
```

Push the tag to remote


```
MINGW64/c/Users/neelamma/Desktop/hospital-system
(Use "git restore <file>..." to discard changes in working directory)
modified: README.md

no changes added to commit (use "git add" and/or "git commit -a")
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git add .
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   README.md

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git tag -a v1.0 -m"Release version 1.0"
fatal: tag 'v1.0' already exists
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git tag -a v2.0 -m"Release version 2.0"
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git tag
v1.0
v2.0
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git show v2.0
tag v2.0
Tagger: VAMSHINEELAMMA <vamshineelamma@gmail.com>
Date: Thu Jan 29 20:39:00 2026 +0530

Release version 2.0

commit 38304ace6a48424b173cd720f718ff9e60379601 (HEAD -> master, tag: v2.0, tag: v1.0, origin/master, origin/HEAD)
Merge: 48a86d7 84c09f4
Author: VAMSHINEELAMMA <vamshineelamma@gmail.com>
Date: Thu Jan 29 19:21:30 2026 +0530

    Merge branch 'feature-login-v1'

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$
```

Assignment 26: Rename Branch

Scenario:

Your team renamed master to main.

Task:

Rename branch locally

Update remote branch

```
MINGW64/c/Users/neelamma/Desktop/hospital-system
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git show v2.0
tag v2.0
Tagger: VAMSHINEELAMMA <vamshineelamma@gmail.com>
Date: Thu Jan 29 20:39:00 2026 +0530

Release version 2.0

commit 38304ace6a48424b173cd720f718ff9e60379601 (HEAD -> master, tag: v2.0, tag: v1.0, origin/master, origin/HEAD)
Merge: 48a86d7 84c09f4
Author: VAMSHINEELAMMA <vamshineelamma@gmail.com>
Date: Thu Jan 29 19:21:30 2026 +0530

    Merge branch 'feature-login-v1'

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git branch
* master
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (master)
$ git branch -m master main
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git push -u origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'main' on GitHub by visiting:
remote:   https://github.com/VAMSHINEELAMMA/hospital-system/pull/new/main
remote:
To https://github.com/VAMSHINEELAMMA/hospital-system.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git push origin --delete master
To https://github.com/VAMSHINEELAMMA/hospital-system.git
! [remote rejected] master (refusing to delete the current branch: refs/heads/master)
error: failed to push some refs to 'https://github.com/VAMSHINEELAMMA/hospital-system.git'

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git push origin --delete master
To https://github.com/VAMSHINEELAMMA/hospital-system.git
- [deleted]        master

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$
```


● Module 7: Real-World Team Scenarios

Assignment 27: Hotfix Branch

Scenario:

A production bug needs immediate fix.

Task:

Create hotfix branch

Fix issue

Merge into main

```
MINGW64/c:/Users/neelamma/Desktop/hospital-system
$ git branch
* main

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git checkout -b hotfix-critical-bug
Switched to a new branch 'hotfix-critical-bug'

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (hotfix-critical-bug)
$ nano README.md

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (hotfix-critical-bug)
$ git add README.md

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (hotfix-critical-bug)
$ git commit -m "hotfix:fixed critical issue"
[hotfix-critical-bug e08ab6f] hotfix:fixed critical issue
1 file changed, 3 insertions(+)

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (hotfix-critical-bug)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git merge hotfix-critical-bug
Updating 38304ac..e08ab6f
Fast-forward
 README.md | 3 +++
1 file changed, 3 insertions(+)

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git branch -d hotfix-critical-bug
Deleted branch hotfix-critical-bug (was e08ab6f).

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git log --oneline --decorate -5
fatal: unrecognized argument: --oneline

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git log --oneline --decorate -5
e08ab6f (HEAD -> main) hotfix:fixed critical issue
38304ac (tag: v2.0, tag: v1.0, origin/main) Merge branch 'feature-login-v1'
48a86d7 docs : update README for master
84c09f4 docs : update README for feature-login-v1
33cc609 feature : add login feature skeleton

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$
```

Assignment 28: Cherry-Pick Commit

Scenario:

You want one specific commit from another branch.

Task:

Cherry-pick the commit into current branch

```
MINGW64/c/Users/neelamma/Desktop/hospital-system
33cc609 feature : add login feature skeleton

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git checkout main
Already on 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git log --oneline --all
e08ab6f (HEAD -> main) hotfix:fixed critical issue
38304ac (tag: v2.0, tag: v1.0, origin/main) Merge branch 'feature-login-v1'
48a86d7 docs : update README for master
84c09f4 docs : update README for feature-login-v1
33cc609 feature : add login feature skeleton
8b069e8 docs:adding more information
4d6579b docs: update README
d82c41f initial commit with README and gitignore

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git cherry-pick 33cc609
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

You are currently cherry-picking commit 33cc609.
(all conflicts fixed: run "git cherry-pick --continue")
(use "git cherry-pick --skip" to skip this patch)
(use "git cherry-pick --abort" to cancel the cherry-pick operation)

nothing to commit, working tree clean
The previous cherry-pick is now empty, possibly due to conflict resolution.
If you wish to commit it anyway, use:

    git commit --allow-empty

Otherwise, please use 'git cherry-pick --skip'

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main|CHERRY-PICKING)
$ git log --oneline -5
e08ab6f (HEAD -> main) hotfix:fixed critical issue
38304ac (tag: v2.0, tag: v1.0, origin/main) Merge branch 'feature-login-v1'
48a86d7 docs : update README for master
84c09f4 docs : update README for feature-login-v1
33cc609 feature : add login feature skeleton

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main|CHERRY-PICKING)
$ |
```

Assignment 29: Clean Repository

Scenario:

Repository has unused files and branches.

Task:

Delete unused local branches

Remove untracked files

```
MINGW64/c/Users/neelamma/Desktop/hospital-system
$ git cherry-pick 33cc609
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

You are currently cherry-picking commit 33cc609.
(all conflicts fixed: run "git cherry-pick --continue")
(use "git cherry-pick --skip" to skip this patch)
(use "git cherry-pick --abort" to cancel the cherry-pick operation)

nothing to commit, working tree clean
The previous cherry-pick is now empty, possibly due to conflict resolution.
If you wish to commit it anyway, use:

    git commit --allow-empty

Otherwise, please use 'git cherry-pick --skip'

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main|CHERRY-PICKING)
$ git log --oneline -5
e08ab6f (HEAD -> main) hotfix:fixed critical issue
38304ac (tag: v2.0, tag: v1.0, origin/main) Merge branch 'feature-login-v1'
48a86d7 docs : update README for master
84c09f4 docs : update README for feature-login-v1
33cc609 feature : add login feature skeleton

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main|CHERRY-PICKING)
$ git branch
* main

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main|CHERRY-PICKING)
$ git cherry-pick --skip

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git branch
* main

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git cherry-pick --abort
error: no cherry-pick or revert in progress
fatal: cherry-pick failed

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git branch
* main

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ |
```

Exit cherry-pick mode

Assignment 30: Rebase Feature Branch

Scenario:

Your feature branch is behind main.

Task:

Rebase feature branch on latest main

Resolve conflicts if any

```
MINGW64/c/Users/neelamma/Desktop/hospital-system
neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (rebase-demo)
$ nano README.md

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (rebase-demo)
$ git add README.md

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (rebase-demo)
$ git commit -m"docs:update README in rebase-demo"
On branch rebase-demo
nothing to commit, working tree clean

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (rebase-demo)
$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(Use "git push" to publish your local commits)

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git add README.md

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git commit -m"docs:update README in rebase-demo"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(Use "git push" to publish your local commits)

nothing to commit, working tree clean

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git commit -m"docs:update README in main"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(Use "git push" to publish your local commits)

nothing to commit, working tree clean

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (main)
$ git checkout rebase-demo
Switched to branch 'rebase-demo'

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (rebase-demo)
$ git rebase main
Current branch rebase-demo is up to date.

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (rebase-demo)
$ git checkout rebase-demo
Already on 'rebase-demo'
```

```
MINGW64/c/Users/neelamma/Desktop/hospital-system

[detached HEAD 107f464] docs:update README in rebase-demo
1 file changed, 4 insertions(+)
Successfully rebased and updated refs/heads/rebase-demo.

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (rebase-demo)
$ git log --oneline --graph
* 107f464 (HEAD -> rebase-demo) docs:update README in rebase-demo
* 744b7cd (main) docs:update README in main
* e08ab6f hotfix:fixed critical issue
* 38304ac (tag: v2.0, tag: v1.0, origin/main) Merge branch 'feature-login-v1'
* 84c09f4 docs : update README for feature-login-v1
* 48a86d7 docs : update README for master
* 33cc609 feature : add login feature skeleton
* 8b069e8 docs:adding more information
* 4d6579b docs: update README
* d52c41f Initial commit with README and gitignore

neelamma@4Gw22AD024 MINGW64 ~/Desktop/hospital-system (rebase-demo)
$ |
```

Bonus Assignment 31: Git Log Analysis

Scenario:

You need to find who changed a specific line.

Task:

Use Git command to identify author and commit